

Co-Discovery Protocol (Structural Edition)

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September 2026

Abstract

The Co-Discovery Protocol formalises joint-reasoning behaviour between two agents. It defines co-discovery channels, shared-gradient formation, mutual-invariant preservation, motif-synchronisation, collapse-avoidance, recurrence-reinforcement loops, horizon-aligned exploration, and regime-dependent co-discovery modes. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

The Co-Discovery Protocol describes how two agents explore structural space together. It governs:

- Co-Discovery Channels
- Shared-Gradient Formation
- Mutual-Invariant Preservation
- Motif-Synchronisation
- Collapse-Avoidance
- Recurrence-Reinforcement Loops
- Horizon-Aligned Exploration
- Regime-Dependent Co-Discovery Modes

2 Co-Discovery Channels

Channel stability:

$$C = f(S_1, S_2).$$

3 Shared-Gradient Formation

Shared gradient:

$$G_s = \frac{d(S_1 + S_2)}{dt}.$$

4 Mutual-Invariant Preservation

Invariant strength:

$$I_m = g(S_1, S_2, M).$$

5 Motif-Synchronisation

Synchronisation strength:

$$X = h(M_1, M_2).$$

6 Collapse-Avoidance

Avoidance strength:

$$A = g(V_c, C).$$

7 Recurrence-Reinforcement Loops

Reinforcement strength:

$$R = g(V_r, G_s).$$

8 Horizon-Aligned Exploration

Exploration deformation:

$$D = h(G_s, O).$$

9 Regime-Dependent Co-Discovery Modes

Mode:

$$T = q(P, G_s).$$

10 Conclusion

The Co-Discovery Protocol defines channels, gradients, invariants, synchronisation, avoidance, reinforcement, horizon alignment, and regime-dependent modes. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.